

Lesson 2-1: Understand Statistical Questions

Connect & apply • 8 tasks • Prepares for Standard 6.SP.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Order 18, 7, 12, 7, 25 from least to greatest.

Answer: 7, 7, 12, 18, 25

Keep both 7s; repeated values still count.

Prerequisite skill: Order a list of up to 10 whole numbers least to greatest

TASK 2 • FOUNDATION 2

A number line goes from 0 to 30 in 6 equal jumps. How much is one jump?

Answer: 5

Divide the total distance by the number of equal jumps: $30 \div 6 = 5$.

Prerequisite skill: Read a scaled number line and say what one tick mark is worth

TASK 3 • FOUNDATION 3

A survey records 4 yes responses, then 5 yes responses, then 3 yes responses. How many yes responses?

Answer: 12

Add each group once: $4 + 5 + 3 = 12$.

Prerequisite skill: Tally and count a set of up to 30 responses without losing place

TASK 4 • FOUNDATION 4

Which is greater: 47 or 74?

Answer: 74

Compare the tens digits: 7 tens is greater than 4 tens.

Prerequisite skill: Compare two whole numbers and say which is greater

Lesson 2-1: Understand Statistical Questions

Connect & apply • 8 tasks • Prepares for Standard 6.SP.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Order 7, 3, 12, 5, 9. On a line from 0 to 10 split into 5 equal jumps, what is one jump worth?

Answer: Ordered list: 3, 5, 7, 9, 12. Line jump value: $10 \div 5 = 2$ units per jump.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Order 15, 8, 22, 11, 4. On a line from 0 to 40 split into 4 equal jumps, what is one jump worth?

Answer: Ordered: 4, 8, 11, 15, 22. One jump = $40 \div 4 = 10$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Order 6, 1, 9, 3, 7. On a line from 0 to 100 split into 10 equal jumps, what is one jump worth?

Answer: Ordered: 1, 3, 6, 7, 9. One jump = $100 \div 10 = 10$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A line runs from 0 to 30 with 6 equal jumps. Where does 25 sit — on a tick mark or between two?

Answer: One jump = $30 \div 6 = 5$, so ticks are 0, 5, 10, 15, 20, 25, 30. The value 25 lands exactly on a tick mark.

Check the result against the situation and explain your reasoning.

Lesson 2-2: Represent and Describe Data in a Histogram

Connect & apply • 8 tasks • Prepares for Standard 6.SP.B.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Which interval contains 29: 10–19, 20–29, or 30–39?

Answer: 20–29

The endpoints are included, so 29 belongs in 20–29.

Prerequisite skill: Decide which interval a value belongs to (is 19 in 10–19 or 20–29?)

TASK 2 • FOUNDATION 2

Sort 2, 8, 11, 14, 19, 22 into intervals 0–9, 10–19, 20–29. Give the three frequencies in order.

Answer: 2, 3, 1

The bins contain (2, 8), (11, 14, 19), and (22). The frequencies sum to 6.

Prerequisite skill: Sort up to 20 values into given intervals and count each bin

TASK 3 • FOUNDATION 3

Find the total frequency: $8 + 12 + 7 + 3$.

Answer: 30

Pair $8 + 12 = 20$ and $7 + 3 = 10$; total 30.

Prerequisite skill: Add a column of one- and two-digit numbers to a total

TASK 4 • FOUNDATION 4

A frequency axis is labeled 0, 5, 10, 15. What value is two labeled steps above 0?

Answer: 10

Each labeled step adds 5, so two steps give 10.

Prerequisite skill: Read a graph scale that skip-counts by 2, 5, or 10

Lesson 2-2: Represent and Describe Data in a Histogram

Connect & apply • 8 tasks • Prepares for Standard 6.SP.B.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Which interval holds 29: 20–29 or 30–39? How many of 4, 11, 19, 21, 25 land in 10–19?

Answer: Interval assignment: 29 belongs in the 20–29 interval. Bin count: Values in 10–19 are 11 and 19 (Total = 2 values).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Which interval holds 40: 30–39 or 40–49? How many of 12, 35, 41, 44, 58 land in 40–49?

Answer: 40 belongs in 40–49. Values in 40–49: 41 and 44 — a frequency of 2.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Which interval holds 60: 50–59 or 60–69? How many of 51, 60, 64, 70, 72 land in 60–69?

Answer: 60 belongs in 60–69. Values in 60–69: 60 and 64 — a frequency of 2.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A data set has 24 values sorted into bins with frequencies 5, 8, 6, and _____. What is the missing frequency?

Answer: $5 + 8 + 6 = 19$, so the missing frequency is $24 - 19 = 5$.

Check the result against the situation and explain your reasoning.

Lesson 2-3: Describe the Data Using the Median

Connect & apply • 8 tasks • Prepares for Standard 6.SP.B.5.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Order 15, 8, 12, 8, 17, 10.

Answer: 8, 8, 10, 12, 15, 17

Include both 8s and arrange all six values.

Prerequisite skill: Order up to 12 numbers least to greatest

TASK 2 • FOUNDATION 2

Find the median of 3, 5, 8, 11, 14.

Answer: 8

The third value has two values on each side.

Prerequisite skill: Find the middle position of an odd-length list by counting in from both ends

TASK 3 • FOUNDATION 3

What number is halfway between 18 and 26?

Answer: 22

The gap is 8; move half the gap, 4, from 18.

Prerequisite skill: Find the number halfway between two numbers

TASK 4 • FOUNDATION 4

Compute $46 \div 2$.

Answer: 23

Split 46 into $40 + 6$, then halve each part: $20 + 3$.

Prerequisite skill: Divide a two-digit sum by 2

Lesson 2-3: Describe the Data Using the Median

Connect & apply • 8 tasks • Prepares for Standard 6.SP.B.5.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Put in order: 9, 4, 15, 6, 11. What number is halfway between 12 and 17?

Answer: Ordered list: 4, 6, 9, 11, 15 (Middle value = 9). Halfway point: $(12 + 17) \div 2 = 29 \div 2 = 14.5$ (or $14 \frac{1}{2}$).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Put in order: 14, 6, 21, 9, 17. What number is halfway between 20 and 25?

Answer: Ordered: 6, 9, 14, 17, 21 (median 14). Halfway: $(20 + 25) \div 2 = 45 \div 2 = 22.5$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Put in order: 30, 12, 25, 18, 21. What number is halfway between 8 and 15?

Answer: Ordered: 12, 18, 21, 25, 30 (median 21). Halfway: $(8 + 15) \div 2 = 23 \div 2 = 11.5$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A set of six values has median 10. The four values 4, 8, 12, 16 are known. Give one possible pair for the other two.

Answer: Many answers. The two middle values must average 10 — for example, adding 2 and 20 gives ordered 2, 4, 8, 12, 16, 20 with median $(8 + 12) \div 2 = 10$.

Check the result against the situation and explain your reasoning.

Lesson 2-4: Represent and Describe Data in a Box Plot

Connect & apply • 8 tasks • Prepares for Standard 6.SP.B.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Find the median of 12, 4, 8, 6, 10.

Answer: 8

Order the list as 4, 6, 8, 10, 12; the middle value is 8.

Prerequisite skill: Order a data set and find its median

TASK 2 • FOUNDATION 2

Split 2, 4, 6, 8, 10, 12 into equal lower and upper halves.

Answer: Lower: 2, 4, 6; upper: 8, 10, 12

There are six values, so put three in each half.

Prerequisite skill: Split an ordered list into a lower half and an upper half

TASK 3 • FOUNDATION 3

Compute $73 - 28$.

Answer: 45

Regroup one ten: $13 - 8 = 5$ and 6 tens $- 2$ tens = 4 tens.

Prerequisite skill: Subtract two-digit whole numbers

TASK 4 • FOUNDATION 4

A line has ticks at 0, 4, 8, 12, 16. Which value is halfway between 8 and 12?

Answer: 10

Half of the interval of 4 is 2; $8 + 2 = 10$.

Prerequisite skill: Plot a point at the right place on a scaled number line

Lesson 2-4: Represent and Describe Data in a Box Plot

Connect & apply • 8 tasks • Prepares for Standard 6.SP.B.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Ordered: 3, 5, 8, 10, 12, 14, 18, 20. Find the median of the lower half.

Answer: Ordered list split: [3, 5, 8, 10] | [12, 14, 18, 20]. Lower half: 3, 5, 8, 10. Median of lower half (Q1): $(5 + 8) \div 2 = 13 \div 2 = 6.5$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Ordered: 2, 6, 9, 11, 15, 19, 22, 28. Find the median of the lower half.

Answer: Lower half: 2, 6, 9, 11. $Q1 = (6 + 9) \div 2 = 15 \div 2 = 7.5$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Ordered: 4, 7, 10, 13, 18, 20. Find the median of the upper half.

Answer: Upper half: 13, 18, 20. $Q3 = 18$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A box plot has min 3, Q1 7, median 12, Q3 18, max 25. Between which two values do the middle half of the data sit?

Answer: Between $Q1 = 7$ and $Q3 = 18$. That middle 50% spans $18 - 7 = 11$ units.

Check the result against the situation and explain your reasoning.

Lesson 2-5: Describe Data by Range and Interquartile Range

Connect & apply • 8 tasks • Prepares for Standard 6.SP.B.5.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Give the minimum and maximum of 18, 7, 24, 11, 16.

Answer: Minimum 7; maximum 24

The minimum is the least value and the maximum is the greatest.

Prerequisite skill: Find the maximum and minimum of a data set

TASK 2 • FOUNDATION 2

Compute $100 - 37$.

Answer: 63

Count up from 37: 3 to 40 and 60 to 100, giving 63.

Prerequisite skill: Subtract whole numbers, including across zeros

TASK 3 • FOUNDATION 3

For 2, 4, 6, 8, 10, 12, 14, 16, find Q1 and Q3 using the medians of the two halves.

Answer: Q1 = 5; Q3 = 13

Lower half median is $(4 + 6) \div 2 = 5$; upper half median is $(12 + 14) \div 2 = 13$.

Prerequisite skill: Locate Q1 and Q3 in an ordered set

TASK 4 • FOUNDATION 4

Set A ranges from 12 to 30. Set B ranges from 8 to 31. Which has the larger range?

Answer: Set B

A has range 18; B has range 23, which is larger.

Prerequisite skill: Compare two differences to say which spread is larger

Lesson 2-5: Describe Data by Range and Interquartile Range

Connect & apply • 8 tasks • Prepares for Standard 6.SP.B.5.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Range of 12, 7, 19, 4, 15? If $Q1 = 6$ and $Q3 = 17$, what is $Q3 - Q1$?

Answer: Range: Maximum – Minimum = $19 - 4 = 15$. IQR: $Q3 - Q1 = 17 - 6 = 11$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Range of 23, 8, 41, 16, 30? If $Q1 = 12$ and $Q3 = 35$, what is $Q3 - Q1$?

Answer: Range = $41 - 8 = 33$. IQR = $35 - 12 = 23$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Range of 100, 62, 85, 71, 94? If $Q1 = 68$ and $Q3 = 92$, what is $Q3 - Q1$?

Answer: Range = $100 - 62 = 38$. IQR = $92 - 68 = 24$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Two classes both have range 20, but Class A has IQR 4 and Class B has IQR 16. What does that tell you?

Answer: The middle half of Class A's data spans 4 units; the middle half of Class B's spans 16 units, so B has the larger spread within its middle half. Both full data sets span 20 units. These summaries alone do not tell us how many outliers there are or how values are distributed within those spans.

Check the result against the situation and explain your reasoning.

Lesson 2-6: Divide Multi-Digit Numbers Using an Algorithm

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.2

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $84 \div 7$.

Answer: 12

$7 \times 12 = 84$, so the quotient is 12.

Prerequisite skill: Recall multiplication and division facts through 12×12

TASK 2 • FOUNDATION 2

Compute $248 \div 4$.

Answer: 62

$24 \text{ tens} \div 4 = 6 \text{ tens}$; $8 \text{ ones} \div 4 = 2 \text{ ones}$.

Combine to get 62.

Prerequisite skill: Use place value to divide hundreds, tens, and ones

TASK 3 • FOUNDATION 3

Compute $302 - 178$.

Answer: 124

Regroup across the zero or count up: $178 + 22 = 200$ and $200 + 102 = 302$; total 124.

Prerequisite skill: Subtract multi-digit whole numbers with regrouping

TASK 4 • FOUNDATION 4

Estimate $238 \div 6$ using the nearby compatible dividend 240.

Answer: 40

$240 \div 6 = 40$. Since 238 is slightly less than 240, the exact quotient will be slightly less than 40.

Prerequisite skill: Estimate a quotient and check with multiplication

Lesson 2-6: Divide Multi-Digit Numbers Using an Algorithm

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.2

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Compute $156 \div 6$. Explain how multiplication checks your quotient.

Answer: $156 \div 6 = 26$. Divide 15 tens by 6: 2 tens with 3 tens remaining. Combine those 3 tens with 6 ones to get 36 ones. Divide 36 by 6 to get 6 ones. Check: $26 \times 6 = 156$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Compute $252 \div 7$. Check with multiplication.

Answer: $252 \div 7 = 36$. Divide 25 tens by 7 to get 3 tens with 4 tens remaining; 42 ones $\div 7 = 6$ ones. Check: $36 \times 7 = 252$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Compute $325 \div 5$. Check with multiplication.

Answer: $325 \div 5 = 65$. Divide 32 tens by 5 to get 6 tens with 2 tens remaining; 25 ones $\div 5 = 5$ ones. Check: $65 \times 5 = 325$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

There are 347 students. Each bus seats at most 24 students. What is the fewest buses needed? Explain how the remainder affects your answer.

Answer: $347 \div 24 = 14$ remainder 11, because $24 \times 14 = 336$. The 11 remaining students need another bus, so 15 buses are needed.

Check the result against the situation and explain your reasoning.

Lesson 2-7: Divide Decimals Using an Algorithm

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $96 \div 8$.

Answer: 12

Use the related multiplication fact: $8 \times 12 = 96$.

Prerequisite skill: Recall division facts through 12×12

TASK 2 • FOUNDATION 2

Compute $156 \div 6$.

Answer: 26

$120 \div 6 = 20$ and $36 \div 6 = 6$; add $20 + 6$.

Prerequisite skill: Divide multi-digit whole numbers with the standard algorithm

TASK 3 • FOUNDATION 3

Compute 4.8×100 .

Answer: 480

Every digit becomes worth 100 times as much.

Prerequisite skill: Multiply and divide by 10, 100, 1000 by shifting the decimal point

TASK 4 • FOUNDATION 4

Compute $7.5 \div 2$.

Answer: 3.75

Write 7.50; divide 750 hundredths by 2 to get 375 hundredths.

Prerequisite skill: Line up decimal places and annex zeros to keep dividing

Lesson 2-7: Divide Decimals Using an Algorithm

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

$84 \div 4$? $4.8 \div 2$? 0.6×100 ?

Answer: $84 \div 4 = 21$ $4.8 \div 2 = 2.4$ $0.6 \times 100 = 60$ (shift decimal 2 places right and annex zero).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$96 \div 3$? $6.9 \div 3$? 0.8×100 ?

Answer: $96 \div 3 = 32$. $6.9 \div 3 = 2.3$. $0.8 \times 100 = 80$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$72 \div 6$? $8.4 \div 4$? 0.05×1000 ?

Answer: $72 \div 6 = 12$. $8.4 \div 4 = 2.1$. $0.05 \times 1000 = 50$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Why does $4.8 \div 0.2$ give the same answer as $48 \div 2$?

Answer: Multiplying both the dividend and the divisor by 10 does not change the quotient; it only rewrites the problem with a whole-number divisor. Both equal 24.

Check the result against the situation and explain your reasoning.

Lesson 2-8: Describe Data Using the Mean

Connect & apply • 8 tasks • Prepares for Standard 6.SP.B.5.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Add $12 + 18 + 15 + 25 + 10$.

Answer: 80

Pair $12 + 18 = 30$ and $15 + 25 = 40$, then add 10.

Prerequisite skill: Add a list of 5–10 two-digit numbers accurately

TASK 2 • FOUNDATION 2

Compute $144 \div 6$.

Answer: 24

$6 \times 24 = 144$; use multiplication to check division.

Prerequisite skill: Divide a three-digit number by a one-digit number

TASK 3 • FOUNDATION 3

How many values are in 4, 4, 7, 9, 9, 9?

Answer: 6

Count every entry, including repeated values.

Prerequisite skill: Count how many values are in a data set

TASK 4 • FOUNDATION 4

Find the missing addend: $18 + 22 + \underline{\quad} = 65$.

Answer: 25

The known addends total 40; $65 - 40 = 25$.

Prerequisite skill: Find a missing addend that reaches a given total

Lesson 2-8: Describe Data Using the Mean

Connect & apply • 8 tasks • Prepares for Standard 6.SP.B.5.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Mean of 10, 14, 8, 12, 16? What must be added to 48 to reach 60?

**Answer: Sum: $10 + 14 + 8 + 12 + 16 = 60$.
Count: 5 values. Mean: $60 \div 5 = 12$. Missing addend: $60 - 48 = 12$.**

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Mean of 20, 18, 25, 17? What must be added to 72 to reach 100?

Answer: Sum = 80, count = 4, mean = $80 \div 4 = 20$. Missing addend: $100 - 72 = 28$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Mean of 9, 15, 6, 12, 18? What must be added to 55 to reach 90?

Answer: Sum = 60, count = 5, mean = $60 \div 5 = 12$. Missing addend: $90 - 55 = 35$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Four scores have a mean of 85. Three of them are 80, 90, and 88. What is the fourth?

Answer: Total must be $4 \times 85 = 340$. The known three sum to 258, so the fourth is $340 - 258 = 82$.

Check the result against the situation and explain your reasoning.

Lesson 2-9: Describe Data by Mean Absolute Deviation

Connect & apply • 8 tasks • Prepares for Standard 6.SP.B.5.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Find the mean of 4, 6, 8, 10.

Answer: 7

The total is 28 and there are 4 values; $28 \div 4 = 7$.

Prerequisite skill: Find the mean of a data set

TASK 2 • FOUNDATION 2

What is the distance between 3 and 11 on a number line?

Answer: 8

Distance is nonnegative: $11 - 3 = 8$.

Prerequisite skill: Subtract to find the distance between two numbers and keep it positive

TASK 3 • FOUNDATION 3

Add the distances 3, 1, 1, 3.

Answer: 8

$3 + 1 + 1 + 3 = 8$.

Prerequisite skill: Add a list of distances

TASK 4 • FOUNDATION 4

Eight units of total deviation are shared across four values. What is $8 \div 4$?

Answer: 2

Divide total deviation by the number of values.

Prerequisite skill: Divide a total by the number of values

Lesson 2-9: Describe Data by Mean Absolute Deviation

Connect & apply • 8 tasks • Prepares for Standard 6.SP.B.5.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

The mean is 20. How far is 26 from the mean?
How far is 14?

**Answer: Distance of 26: $|26 - 20| = 6$.
Distance of 14: $|14 - 20| = 6$. (Both distances are strictly 6 units).**

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

The mean is 30. How far is 37 from the mean?
How far is 22?

Answer: $|37 - 30| = 7$. $|22 - 30| = 8$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

The mean is 50. How far is 50 from the mean?
How far is 43?

Answer: $|50 - 50| = 0$. $|43 - 50| = 7$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A set has mean 10 and MAD 0. What must be true of the data?

Answer: Every value equals 10. A MAD of zero means no value deviates from the mean at all.

Check the result against the situation and explain your reasoning.

Lesson 2-10: Choose Appropriate Measures

Connect & apply • 8 tasks • Prepares for Standard 6.SP.B.5.d

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

For 2, 4, 6, 8, find mean, median, range, IQR, and MAD. Use medians of the two halves for quartiles.

Answer: Mean 5; median 5; range 6; IQR 4; MAD 2

Mean: $20 \div 4$. Median: $(4 + 6) \div 2$. Range: $8 - 2$.
Q1 = 3, Q3 = 7. Distances from 5 are 3, 1, 1, 3; their mean is 2.

Prerequisite skill: Compute mean, median, range, IQR, and MAD on demand

TASK 2 • FOUNDATION 2

Which value is a likely outlier: 8, 9, 10, 11, 52?

Answer: 52

Four values are close together; 52 is far from that cluster.

Prerequisite skill: Spot an outlier in a short list

TASK 3 • FOUNDATION 3

A dot plot has 1 dot at 1, 2 dots at 2, 3 dots at 3, 2 dots at 4, and 1 dot at 5. Is it symmetric or skewed?

Answer: Symmetric

Frequencies mirror each other around 3.

Prerequisite skill: Read a dot plot or histogram as symmetric or skewed

TASK 4 • FOUNDATION 4

Replace 6 with 60 in 2, 4, 6. Which changes more: mean or median?

Answer: Mean

The median stays 4. The mean changes from 4 to 22.

Prerequisite skill: Compare two measures and say which is pulled by an extreme value

Lesson 2-10: Choose Appropriate Measures

Connect & apply • 8 tasks • Prepares for Standard 6.SP.B.5.d

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Which measure of center fits 2, 3, 3, 4, 50 — mean or median? Why?

Answer: Best Measure: Median (3). Why: 50 is an extreme outlier that skews the data. The mean is $62 \div 5 = 12.4$, which is higher than 4 out of the 5 data values and does not represent the center. The median (3) accurately reflects the typical value.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Which measure of center fits 5, 6, 6, 7, 80 — mean or median? Why?

Answer: Median (6). The mean is $104 \div 5 = 20.8$, which is larger than four of the five values because 80 pulls it up.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Which measure of center fits 41, 44, 45, 46, 49 — mean or median? Why?

Answer: Either works. The data are symmetric with no outlier; the mean (45) and median (45) agree.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Give a data set of five values where mean and median are equal but the range is 40.

Answer: Many answers. For example 10, 25, 30, 35, 50: mean = $150 \div 5 = 30$, median = 30, range = $50 - 10 = 40$.

Check the result against the situation and explain your reasoning.